| | | | | |
|---|---|---|---|---|
| 2 | No Specific/无特殊要求 | 1 | Gun - Heat, 80-260℃ (180-500℉)<br>80-260℃ (180-500 ℉)热风枪 | - Remark/备注: Refer to Table 4 Heat gun details.<br>- Eff/适用性: ALL<br>- From/来源: C919-A-73-21-45-01A-25BA-A |
| 3 | RC331931 | 1 | Vacuum pump kit<br>真空泵套装 | - Remark/备注: The vacuum pump kit is supplied by Rhinestahl or equivalent as specified in Table 3 Vacuum pump kit details. Follow the instructions of the manufacturer for the operation of the vacuum pump.<br>- Eff/适用性: ALL |
| 4 | No Specific/无特殊要求 | As Req/按需 | Air Source - Compressed, Regulated, 100psi<br>100psi压缩空气源 | - Eff/适用性: ALL |
| 5 | No Specific/无特殊要求 | As Req/按需 | Air Source - Nitrogen, Regulated, 100psi<br>100psi氮气源 | - Eff/适用性: ALL |
| 6 | No Specific/无特殊要求 | 1 | Syringe<br>注射器 | - Remark/备注: Material is polyethylene or equivalent, minimum capacity of 3 ml, maximum diameter of 5 mm, and minimum length of 50 mm.<br>- Eff/适用性: ALL |
| 7 | No Specific/无特殊要求 | As Req/按需 | Torque wrench<br>力矩扳手 | - Remark/备注: range to 30.5 N.m~33.8 N.m(270 lbf.in~300 lbf.in) 范围30.5 N.m~33.8 N.m(270 lbf.in~300 lbf.in)<br>- Eff/适用性: ALL |

**Parts & Material 部件和航材:**

| | | | | |
|---|---|---|---|---|
| | Not Required<br>不需要 | | | |

**Work Content 工作内容:**

ALL

TASK C919-A-73-21-45-01A-25BA-A

Pressure sub-system (and sense line dry out items)-Cleaning
压力子系统(和传感管干燥项目)-清洁

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

警告: 当发动机停止运转后对发动机部件进行工作时要小心。使用合适的隔热手套。发动机部件会在关车后保持高温一小时并可能会烫伤工作者。

WARNING: PUT A WARNING NOTICE(S) TO TELL PERSONS NOT TO START THE ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE ENGINE OR EQUIPMENT.

警告: 放置一个警告牌，告知他人不要起动发动机。如果不遵守此指引，会造成人员受伤和发动机或设备损坏。

WARNING: REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF CONSUMABLE PRODUCTS.

警告: 在危险、存储、安全处理和正常使用消耗产品时参考产品标签和厂家安全数据单SDS的指导。

WARNING: YOU MUST DEACTIVATE THE THRUST REVERSER BEFORE YOU DO WORK ON OR AROUND IT. IF YOU DO NOT, THE THRUST REVERSER CAN OPERATE ACCIDENTALLY AND CAUSE AN INJURY AND/OR DAMAGE.

警告: 在反推上或者在反推附件执行工作之前，必须限动反推。否则，反推可能突然作动并造成受伤或破坏。

WARNING: PUT A WARNING NOTICE(S) TO TELL PERSONS NOT TO ENERGIZE THE FADEC L(R). IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE INJURIES TO PERSONNEL.

警告: 放置一个警告牌，告知他人不要给FADEC L(R)通电。如果不遵守此指引，可能会造成人员受伤。

WARNING: USE PROTECTIVE GOGGLES AND GLOVES WHEN YOU REMOVE OR INSTALL LOCKWIRE. EACH TIME YOU CUT LOCKWIRE, REMOVE AND DISCARD IT IMMEDIATELY. LOOSE LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND, AND/OR CAN CAUSE DAMAGE.

警告: 拆下或安装保险丝时，佩戴护目镜和手套。每次剪断保险丝后，立即拆下并报废它。松动的保险丝会划伤或使人失明，并/或可能造成损伤。

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND LOCAL REGULATIONS.

警告: 使用消耗材料时要小心。遵守材料供应商的说明指示和当地规章要求。

CAUTION: DO NOT APPLY MORE HEAT THAN NECESSARY TO THE COMPONENT. TOO MUCH HEAT WILL CAUSE DAMAGE TO THE COMPONENT.

注意: 不要对部件过度加热。过度加热会损伤部件。

CAUTION: INSTALL PROTECTIVE COVERS AND PROTECTIVE CAPS ON ALL OPEN PORTS, OPEN TUBES, OPEN HOSES AND CONNECTORS AS YOU DISCONNECT THEM TO AVOID DAMAGE TO THE EQUIPMENT.

注意: 当断开时，在所有开口、管开口、软管开口和连接器开口安装保护盖和保护帽，以避免损伤设备。

CAUTION: BEFORE INSTALLATION, MAKE SURE THAT ALL MATING FACES ARE CLEAN AND HAVE NO CONTAMINANTS. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE ENGINE OR EQUIPMENT.

注意： 安装前，确保所有配合面清洁且没有污染物。如果不遵循此指引，可能会造成发动机或设备受损。

CAUTION: ALWAYS OBEY MANUFACTURER'S INSTRUCTIONS WHEN YOU OPERATE EQUIPMENT. DAMAGE TO EQUIPMENT CAN OCCUR.

注意： 操作设备时，全程遵守制造商的指引。可能会损坏设备。

CAUTION: FOR PRE SB 73-0006 PSS, USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUT. USE ONE WRENCH TO HOLD THE FITTING AND THE OTHER TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

注意： 对于未执行SB 73-0006的PSS，使用两个扳手松开或拧紧连接螺帽。使用其中的一个扳手来固定接头，另一个松开或拧紧连接螺帽。否则，会导致设备损坏。

NOTE: For POST SB 73-0006 PSS, two wrenches are not necessary. Only one wrench must be used to loosen or tighten the coupling nut.

注： 对于执行了SB 73-0006的PSS，无需使用两个扳手。仅需使用一个扳手来松开或拧紧连接螺帽。

1. Procedures
主程序

A. Job Set-up
工作准备

(1) On the center pedestal:
在中央操纵台：

(a) Make sure that L(R) FUEL CONTROL switch was in OFF position not less than ten minutes before you do this procedure.
执行本程序前，确保L(R) FUEL CONTROL电门在OFF位置不小于十分钟。

(b) Make sure that L(R) START/IGNITION switch was in NORM position not less than ten minutes before you do this procedure.
执行本程序前，确保L(R) START/IGNITION 电门在NORM位置不小于十分钟。

(c) Put the WARNING NOTICE(S) in position to tell personnel not to operate:
将警告牌放置就位以告知其他人不要操作：

-The L(R) FUEL CONTROL switch.
L(R) FUEL CONTROL电门

-The L(R) START/IGNITION switch.
L(R) START/IGNITION 电门

(2) On the overhead panel:
在头顶面板上：

(a) Make sure that the ON legend of the L(R) FADEC MAINT POWER switch is released (ON legend is off).
确保L(R) FADEC MAINT POWER 电门松开（ON灯熄灭）。

(b) Put a WARNING NOTICE on the FADEC Maintenance Control panel to tell personnel not to operate the L(R) FADEC MAINT POWER switch.
把警告标牌放置到FADEC维护面板上以告知他人不要操作L(R) FADEC MAINT POWER 电门。

(3) Get access to fan area. Refer to Power plant (Fan area)-Open for access procedure, MPP C919-A-71-00-00-03A-540A-A.
接近风扇区域。参考：动力装置（风扇区域） – 为接近打开程序,MPP C919-A-71-00-00-03A-540A-A。

B. Procedure
步骤

(1) For the procedure without the optional parts given inTable Table1 Optional parts , doStep Step B.(3). thruStep Step B.(8). ,Step Step B.(11). andStep Step B.(13). .
对于没有在表1中给出的可选部件的程序，执行步骤B.(3)至步骤B.(8)，步骤B.(11)和步骤B.(13)。

(2) For the procedure with the optional parts given inTable Table1 Optional parts , doStep Step B.(3). thruStep Step B.(4). andStep Step B.(9). thruStep Step B.(13). .
对于有表1中给出的可选部件的程序，执行步骤B.(3)至步骤B.(4)，步骤B.(9)至步骤B.(13)。

NOTE: If you use the optional parts, the PS3 pressure port [13] and P3B hose pressure port [14] on the PSS [1] can be vacuumed together and the heat source must be applied to the PS3 pressure port [13] and P3B hose pressure port [14].

注: 如果使用可选部件， PSS [1]上的 PS3 压力端口[13]和 P3B 软管压力端口[14]可以一起抽真空，并且热源必须应用于 PS3 压力口[13]和 P3B 软管压力口[14]。

**Table 1 Optional parts**
**表1 可选部件**

| PART NUMBER<br>件号 | NOMENCLATURE<br>名称 | QUANTITY<br>数量 |
|---|---|---|
| RC331931P21 | Teflon-6 SST Braided Teflon Hose Assembly (18 inches W/FJIC E/E)<br>特氟龙-6 SST编织特氟龙软管组件(18英寸W/FJIC E/E) | 2 |
| RC331931P22 | J6T Union Tee SS Fitting<br>J6T管接头三通SS接头 | 1 |

**CAUTION: DO NOT OVERLOAD THE PS3 SENSE LINE.**
**注意: 不要使PS3感应管过载。**

(3) Disconnect the PS3 sense line [ 2 ] from the PSS [ 1 ], loose clamp(s) and/or bracket(s) as necessary. Refer to PS3 line- Removal,MPP C919-A-73-21-70-01A-520A-A .
从PSS[1]上断开PS3传感管[2]，按需松开卡箍和/或支架。参考PS3管 – 拆下， MPP C919-A-73-21-70-01A-520A-A,。

**CAUTION: DO NOT OVERLOAD THE P3B SENSE LINE.**
**注意: 不要使P3B感应管过载。**

(4) Disconnect the P3B sense line [ 3 ] from the PSS [1], loose clamp(s) and/or bracket(s) as necessary. Refer to P3B line- Removal, MPP C919-A-73-21-75-01A-520A-A.
从PSS[1]上断开P3B传感管[3],按需松开卡箍和/或支架。参考P3B管 - 拆下, MPP C919-A-73-21-75-01A-520A-A。

(5) Do the preparation for the removal of the moisture from the PS3 [4] (without optional parts) as follows:
按以下步骤，执行PS3[4](无选装部件)除水的准备工作：

**CAUTION: REMOVE AS MUCH STANDING WATER FROM THE PS3 PRESSURE PORT AS POSSIBLE BEFORE YOU BEGIN THE VACUUM DRYING PROCEDURE. IF YOU DO NOT REMOVE THE STANDING WATER, IT WILL DEGRADE THE CAPABILITY OF THE VACUUM PUMP AND CA N DAMAGE EQUIPMENT.**

**注意： 在开始真空干燥程序之前，尽可能多地从PS3压力口中去除积水。如果不清除积水，会降低真空泵的性能，并可能损坏设备。**

(a) After you get access to the PS3 pressure port [13], put a Pipette or Syringe into th e opening of the PS3 [4] transducer and remove the visual standing water that yo u can see. Continue the water extraction until you cannot extract more water. Refe r toFigure Refer to Fig.2 Connection of vacuum pump tool .
接近PS3压力口[13]后，将移液器或注射器放入PS3[4]传感器的开口中，并清除可见的积水。持续吸水，直到无法吸出更多的水为止。参考图2. 真空泵工具的连接。

**CAUTION: MAKE SURE THAT YOU CHANGE THE OIL IN THE VACUUM PUMP A FTER 50 USES OR THE VACUUM PUMP EFFECTIVENESS CAN BE C OMPROMISED.**

**注意： 确保在使用50次后更换真空泵中的滑油，否则可能会影响真空泵的效率。**

(b) Connect the vacuum hose [12] end adapter of the vacuum pump [11] to the PS3 pr essure port [13] on the PSS [1]. TORQUE the fitting to 30.5 N.m ~ 33.8 N.m( 270 lbf.in ~ 300 lbf.in).
将真空泵[11]的真空软管[12]转接头连接到PSS[1]上的PS3压力端[13]。磅紧接头到 30.5 N.m ～ 33.8 N.m ( 270 lbf.in ～ 300 lbf.in )。

**CAUTION: IF A NON-VARIABLE HEAT GUN IS USED WHEN AMBIENT TEMPER ATURE IS BELOW 65 DEGF (18.3 DEGC), IT CAN RESULT IN AN IN COMPLETE DRYOUT.**

**注意： 如果在环境温度低于65华氏度（18.3摄氏度）时使用不可调热风枪，可能会导致干燥不彻底。**

(c) If the ambient temperature is below 65 ℉ 18.3 ℃ ), use a Gun - Heat, 80-260℃ (180-500 ℉) to heat the PS3 pressure port [13].
如果环境温度低于 65 ℉ 18.3 ℃ ), 使用80-260℃(180-500℉)热风枪来加热PS3压力口[13]。

-Refer toTable to set the Gun - Heat, 80-260℃ (180-500 ℉) temperature.
**参考表2热风枪温度表来设置80-260℃(180-500℉)热风枪的温度。**

-Apply heat to the PS3 pressure port [13] with the Gun - Heat, 80-260℃ (180-500 ℉) for 4 minutes before you start the vacuum dry-out procedure and apply the Gun - Heat, 80-260℃ (180-500 ℉) until the procedure is completed. Refer toFigure Refer to Fig.1 PSS , PS3 sense line, and P3B sense line .

开始真空干燥程序前使用80-260℃(180-500℉)热风枪给PS3压力口[13]加热4分钟，并用80-260℃(180-500℉)热风枪加热直到程序结束。参考图1.PSS、PS3传感管和P3B传感管。

-AMBIENT TEMPERATURE
环境温度

-MAXIMUM HEAT GUN SETTING
最大热风枪温度设置

-65 ℉ 18.3 ℃ ) to 50 ℉ 10.0 ℃ ).
65℉(18.3℃)至50℉(10.0℃).

-250 ℉ ( 121.1 ℃ ).

-49 ℉ 9.4 ℃ ) to 32 ℉ 0.0 ℃ ).
49℉(9.4 ℃)至32 ℉(0.0 ℃ ).

-300 ℉ ( 148.9 ℃ ).

-31 ℉ 0.5 ℃ ) to 14 ℉ 10.0 ℃ ).
31 ℉(-0.5 ℃)至 14 ℉(-10.0 ℃ ).

-400 ℉ ( 204.4 ℃ ).

-Below 13 ℉ 10.5 ℃ .
低于 13 ℉(-10.5 ℃) .

-500 ℉ ( 260.0 ℃ ).

(6) Remove the moisture from the PS3 [4] (without optional parts) as follows:
按以下步骤，执行PS3[4](无选装部件)除水：

(a) Turn the vacuum relief valve [9] and blank-off valve [10] handles to the open positi on.
将真空释压活门[9]和断开活门[10]手柄转动到打开位。

**CAUTION: DO NOT LET CONTAMINATION BE INGESTED INTO THE VACUUM P UMP RELIEF PORT. IF THE PRESSURE READING IS NOT BELOW TH E SPECIFIED THRESHOLD, THE PROCEDURE COULD NOT BE EFFE CTIVE.**
**注意： 不要让污染物被吸入到真空泵释压口。如果压力读数不低于规定的阈值，该程序可能无效。**

(b) Start the vacuum pump [11] and let the pump motor get to the operating speed.
启动真空泵[11]并让泵马达达到工作速度。

**CAUTION: DO NOT CLOSE THE BLANK-OFF VALVE WHILE YOU REMOVE MO ISTURE FROM THE PSS. IF THE BLANK-OFF VALVE IS CLOSED, M OISTURE WILL NOT BE REMOVED.**

**注意：　　当从PSS去除水分时不要关闭断开活门。否则，水分将不会被去除。**

(c) Turn the valve handle of the vacuum relief valve [9] to the horizontal position to clo se it.
转动真空释压活门[9]的手柄到水平位置以关闭它。

(d) Make sure that sufficient vacuum is pulled on the PS3 [4] manifold by the vacuum pump [11].
确保真空泵[11]在PS3[4]总管上抽到足够的真空。

NOTE: The vacuum read needs to be corrected based on the pressure altitude at which the pump is used. Refer toFigure Refer to F ig.3 Vacuum gauge corrections for altitude .

**注：** 真空读数需要基于泵在使用的压力海拔高度进行修正。参考图3.真空表海拔高度修正。

- If the airport elevation is below 5000 ft ( 1524 m ), make sure that the vacuum measurement is in the green-colored region on the vacuum pressure indicator.
如果机场海拔低于 5000 ft ( 1524 m )，确保真空测量值在真空压力指示器的绿色区域内。

- If the airport elevation is above 5000 ft ( 1524 m ), refer toFigure Refer to Fig.3 Vacuum gauge corrections for altitude to find the correct vacuum pressure meas urement.
-如果机场海拔高于 5000 ft ( 1524 m )，参考图3.真空表海拔高度修正，以找到正确的真空压力测量值。

- If the vacuum measurement is not correct, examine for leaks in the vacuum ho se, connections, and make sure the pump valves are in the correct position.
-如果真空测量值不正确，检查真空软管、连接是否渗漏，并确保泵活门处于正确位置。

(e) Operate the vacuum pump [11] for a minimum of 20 minutes.
真空泵[11]运行至少20分钟。

(f) After 20 minutes of operation, open the vacuum relief valve [9] to release the press ure.
运行20分钟后，打开真空释压活门[9]以释放压力。

(g) Turn OFF the vacuum pump [11].
关闭真空泵[11]。

(h) Remove the vacuum hose [12] end adapter from the PS3 pressure port [13].
从PS3压力口[13]拆下真空管[12]端转接头。

(i) Blow clean dry Air Source-Compressed,Regulated,100 psi or Air Source - Nitroge n, Regulated, 100 psi (no more than 200 psi ( 1379 kPa )) in the PS3 sense line [ 2 ] for a minimum of 3 minutes. Continue until no moisture comes out from the PS 3 sense line [ 2 ] weep hole located at the 4:00 o'clock position in the aft location of the fan compartment. Refer toFigure Refer to Fig.4 Location of Sense Line Weep Holes .
向PS3传感管[2]中吹清洁、干燥的100psi可调压缩空气源或100 psi可调氮气源(不超过 200 psi (1379 kPa ))至少3分钟。继续直到没有水汽从位于风扇舱后部4:00点钟位置的PS3传感管[2]渗水孔流出。参考图4.传感管渗水孔位置。

(j) Make sure that the PS3 sense line [2] weep hole is not blocked because of ice formation, damage, or contamination, and that there is a free flow of air from the PS3 sense line [2] weep hole.
确保PS3传感管[2]渗水孔没有因结冰、损伤或污染而堵塞，并且有空气从PS3传感管[2]的渗水孔自由流出。

NOTE: For best results, compressed Air Source - Nitrogen, Regulated, 100 psi is recommended.

**注：** 为了获得最佳效果，建议使用100 psi可调氮气源。

(7) Do the preparation for the removal of the moisture from the P3B [5] (without optional parts) as follows:
按以下步骤，执行P3B[5]（无选装部件）除水的准备工作：

**CAUTION: REMOVE AS MUCH STANDING WATER FROM THE P3B PRESSURE PORT AS POSSIBLE BEFORE YOU BEGIN THE VACUUM DRYING PROCEDURE. IF YOU DO NOT REMOVE THE STANDING WATER, IT WILL DEGRADE THE CAPABILITY OF THE VACUUM PUMP AND CAN DAMAGE EQUIPMENT.**

**注意： 在开始真空干燥程序之前，尽可能多地从P3B压力口中去除积水。如果不清除积水，会降低真空泵的性能，并可能损坏设备。**

(a) After you get access to the P3B hose pressure port [14], put a Pipette or Syringe into the open port of the of the P3B [5] transducer and remove the visible standing water that you can see. Continue the water extraction until you cannot extract more water. Refer toFigure  Refer to Fig.2 Connection of vacuum pump tool .
接近P3B压力口[14]后，将移液器或注射器放入P3B[5]传感器的开口中，并清除可见的积水。持续吸水，直到无法吸出更多的水为止。参考图2. 真空泵工具的连接。

**CAUTION: MAKE SURE THAT YOU CHANGE THE OIL IN THE VACUUM PUMP AFTER 50 USES OR THE VACUUM PUMP EFFECTIVENESS CAN BE COMPROMISED.**

**注意： 确保在使用50次后更换真空泵中的滑油，否则可能会影响真空泵的效率。**

(b) Connect the vacuum hose [12] end adapter of the vacuum pump [11] to the P3B hose pressure port [14] on the PSS [1]. TORQUE the fitting to 30.5 N.m ~ 33.8 N.m( 270 lbf.in ~ 300 lbf.in).
将真空泵[11]的真空软管[12]转接头连接到PSS[1]上的P3B软管压力口[14]。磅紧接头 到 30.5 N.m ～ 33.8 N.m ( 270 lbf.in ～ 300 lbf.in )。

**CAUTION: IF A NON-VARIABLE HEAT GUN IS USED WHEN AMBIENT TEMPERATURE IS BELOW 65 DEGF (18.3 DEGC), IT CAN RESULT IN AN INCOMPLETE DRYOUT.**

**注意： 如果在环境温度低于65华氏度（18.3摄氏度）时使用不可调热风枪，可能会导致干燥不彻底。**

(c) If the ambient temperature is below 65 ℉ 18.3 ℃ ), use a Gun - Heat, 80-260℃ (180-500 ℉) to heat the P3B hose pressure port [14].
如果环境温度低于 65 ℉(18.3 ℃ )，使用80-260℃(180-500 ℉)热风枪来加热P3B软管压力口[14]。

-Refer toTable  to set the Gun - Heat, 80-260℃ (180-500 ℉) temperature.
**参考表2热风枪温度表来设置80-260℃(180-500℉)热风枪的温度。**

-Apply heat to the P3B hose pressure port [14] with the Gun - Heat, 80-260℃ (180-500 ℉) for 4 minutes before you start the vacuum dry-out procedure and apply the Gun - Heat, 80-260℃ (180-500 ℉) until the procedure is completed. Refer toFigure Refer to Fig.1 PSS , PS3 sense line, and P3B sense line .

**开始真空干燥程序前使用80-260℃(180-500℉)热风枪给P3B软管压力口[14]加热4分钟，并用80-260℃(180-500℉)热风枪加热直到程序结束。参考图1.PSS、PS3传感管和P3B传感管。**

(8) Remove the moisture from the P3B [5] (without optional parts) as follows:
按以下步骤，执行P3B[5](无选装零件)除水：

(a) Turn the vacuum relief valve [9] and blank-off valve [10] handles to the open position.
将真空释压活门[9]和断开活门[10]手柄转动到打开位。

**CAUTION: DO NOT LET CONTAMINATION BE INGESTED INTO THE VACUUM PUMP RELIEF PORT. IF THE PRESSURE READING IS NOT BELOW THE SPECIFIED THRESHOLD, THE PROCEDURE COULD NOT BE EFFECTIVE.**

**注意: 不要让污染物被吸入到真空泵释压口。如果压力读数不低于规定的阈值，该程序可能无效。**

(b) Start the vacuum pump [11] and let the pump motor get to the operating speed.
启动真空泵[11]并让泵马达达到工作速度。

**CAUTION: DO NOT CLOSE THE BLANK-OFF VALVE WHILE YOU REMOVE MOISTURE FROM THE PSS. IF THE BLANK-OFF VALVE IS CLOSED, MOISTURE WILL NOT BE REMOVED.**

**注意: 当从PSS去除水分时不要关闭断开活门。否则，水分将不会被去除。**

(c) Turn the valve handle of the vacuum relief valve [9] to the horizontal position to close it.
转动真空释压活门[9]的手柄到水平位置以关闭它。

(d) Make sure that sufficient vacuum is pulled on the P3B [5] manifold by the vacuum pump [11].
确保真空泵[11]在P3B[5]总管上抽到足够的真空。

NOTE: The vacuum read needs to be corrected based on the pressure altitude at which the pump is used. Refer toFigure Refer to Fig.3 Vacuum gauge corrections for altitude .

**注:** 真空读数需要基于泵在使用的压力海拔高度进行修正。参考图3.真空表海拔高度修正。

- If the airport elevation is below 5000 ft ( 1524 m ), make sure that the vacuum measurement is in the green-colored region on the vacuum pressure indicator.
如果机场海拔低于 5000 ft ( 1524 m )，确保真空测量值在真空压力指示器的绿色区域内。

- If the airport elevation is above 5000 ft ( 1524 m ), refer toFigure Refer to Fig.3 Vacuum gauge corrections for altitude to find the correct vacuum pressure measurement.
-如果机场海拔高于 5000 ft ( 1524 m )，参考图3.真空表海拔高度修正，以找到正确的真空压力测量值。

- If the vacuum measurement is not correct, examine for leaks in the vacuum hose, connections, and make sure the pump valves are in the correct position.
-如果真空测量值不正确，检查真空软管、连接是否渗漏，并确保泵活门处于正确位置。

(e) Operate the vacuum pump [11] for a minimum of 20 minutes.
真空泵[11]运行至少20分钟。

(f) After 20 minutes of operation, open the vacuum relief valve [9] to release the pressure.
运行20分钟后，打开真空释压活门[9]以释放压力。

(g) Turn OFF the vacuum pump [11].
关闭真空泵[11]。

(h) Remove the vacuum hose [12] end adapter from the P3B hose pressure port [14].
从P3B软管压力口[14]拆下真空管[12]端转接头。

(i) Blow clean dry Air Source-Compressed,Regulated,100 psi or Air Source - Nitrogen, Regulated, 100 psi (no more than 200 psi ( 1379 kPa )) in the P3B sense line [ 3 ] for a minimum of 3 minutes. Continue until no moisture comes out from the P3B sense line [ 3 ] weep hole located at the 4:00 o'clock position in the aft location of the fan compartment. Refer toFigure Refer to Fig.4 Location of Sense Line Weep Holes .
向P3B传感管[3]中吹清洁、干燥的100psi可调压缩空气源或100 psi可调氮气源(不超过 200 psi(1379 kPa ))至少3分钟。继续直到没有水汽从位于风扇舱后部4:00点钟位置的P3B传感管[3]渗水孔流出。参考图4.传感管渗水孔位置。

(j) Make sure that the P3B sense line [3] weep hole is not blocked because of ice formation, damage, or contamination, and that there is a free flow of air from the P3B sense line [3] weep hole.
确保P3B传感管[3]渗水孔没有因结冰、损伤或污染而堵塞，并且有空气从P3B传感管[3]的渗水孔自由流出。

NOTE: For best results, compressed Air Source - Nitrogen, Regulated, 100 psi is recommended.
**注:** 为了获得最佳效果，建议使用100 psi可调氮气源。

(9) Do the preparation for the removal of the moisture from the PS3 [4] and P3B [5] (with optional parts) as follows:
按以下步骤，执行PS3[4]和P3B[5](有选装零件)除水的准备工作:

**CAUTION: REMOVE AS MUCH STANDING WATER FROM THE PS3 AND P3B PRESSURE PORTS AS POSSIBLE BEFORE YOU BEGIN THE VACUUM DRYING PROCEDURE. IF YOU DO NOT REMOVE THE STANDING WATER, IT WILL DEGRADE THE CAPABILITY OF THE VACUUM PUMP, WHICH COULD CAUSE DAMAGE TO THE EQUIPMENT.**
**注意: 在开始真空干燥程序之前，尽可能多地从PS3和P3B压力口中去除积水。如果不清除积水，会降低真空泵的性能，可能损坏设备。**

(a) After you get access to the PS3 pressure port [13] and P3B hose pressure port [14], put a Pipette or Syringe into the open port of the of the PS3 [4] and P3B [5] transducers and remove the visible standing water that you can see. Continue the water extraction until you cannot extract more water. Refer toFigure Refer to Fig.2 Connection of vacuum pump tool .
接近PS3压力口[13]和P3B软管压力口[14]后，将移液器或注射器放入PS3[4]和P3B[5]的传感器的开口中，并清除可见的积水。持续吸水，直到无法吸出更多的水为止。参考图2. 真空泵工具的连接。

**CAUTION: MAKE SURE THAT YOU CHANGE THE OIL IN THE VACUUM PUMP AFTER 50 USES OR THE VACUUM PUMP EFFECTIVENESS CAN BE COMPROMISED.**

**注意： 确保在使用50次后更换真空泵中的滑油，否则可能会影响真空泵的效率。**

(b) Connect one teflon hose [15] of the Vacuum pump kit(RC331931) to the P3B hose pressure port [14] on the PSS [1]. Refer toFigure Refer to Fig.2 Connection of vacuum pump tool .
将真空泵(RC331931)的一根特氟龙软管[15]连接到PSS[1]上的P3B软管压力口[14]。参考图2. 真空泵工具的连接。

(c) Connect the other teflon hose [15] of the Vacuum pump kit(RC331931) to the PS3 pressure port [13] on the PSS [1]. Refer toFigure Refer to Fig.2 Connection of vacuum pump tool .
将真空泵(RC331931)的另一根特氟龙软管[15]连接到PSS[1]上的PS3压力口[13]。参考图2. 真空泵工具的连接。

(d) Connect the loose ends of each teflon hose [15] to the two horizontal ports of the union tee [16] of the Vacuum pump kit(RC331931) . Refer toFigure Refer to Fig.2 Connection of vacuum pump tool .
将每个特氟龙软管[15]的松端连接到真空泵套件(RC331931)的三通接头[16]的两个水平管口。参考图2. 真空泵工具的连接。

(e) Connect the vacuum hose [12] of the Vacuum pump kit(RC331931) to the T-end of the union tee [16] of the Vacuum pump kit(RC331931) . Refer toFigure Refer to Fig.2 Connection of vacuum pump tool .
将真空泵套件(RC331931)的真空软管[12]连接到真空泵套件(RC331931)的三通接头[16]的T字管口。参考图2. 真空泵工具的连接。

(f) TORQUE all fittings to 30.5 N.m ~ 33.8 N.m( 270 lbf.in ~ 300 lbf.in).
磅紧所有接头 到 30.5 N.m ～ 33.8 N.m ( 270 lbf.in ～ 300 lbf.in )。

The torque/力矩： ______ ☐磅英寸 (lbf.in) ☐牛米 (N.m)

CTC NO./计量器具编号： ______

**CAUTION: IF A NON-VARIABLE HEAT GUN IS USED WHEN AMBIENT TEMPERATURE IS BELOW 65 DEGF (18.3 DEGC), IT CAN RESULT IN AN INCOMPLETE DRYOUT.**

**注意： 如果在环境温度低于65华氏度（18.3摄氏度）时使用不可调热风枪，可能会导致干燥不彻底。**

(g) If the ambient temperature is below 65 ℉ 18.3 ℃ ), use a Gun - Heat, 80-260℃ (180-500 ℉) to heat the PS3 pressure port [13] and P3B hose pressure port [14].
如果环境温度低于 65 ℉(18.3 ℃ )，使用80-260℃(180-500 ℉)热风枪来加热PS3压力口[13]和P3B软管压力口[14]。

-Refer toTable  to set the Gun - Heat, 80-260℃ (180-500 ℉) temperature.
参考表2热风枪温度表来设置80-260℃(180-500℉)热风枪的温度。

-Apply heat to the PS3 pressure port [13] and P3B hose pressure port [14] with the Gun - Heat, 80-260℃ (180-500 ℉) for 4 minutes before you start the vacuum dry-out procedure and apply the Gun - Heat, 80-260℃ (180-500 ℉) until the procedure is completed. Refer toFigure Refer to Fig.1 PSS , PS3 sense line, and P3B sense line .
**开始真空干燥程序前使用80-260℃(180-500℉)热风枪给PS3压力口 [13]和P3B软管压力口[14]加热4分钟，并用80-260℃(180-500℉)热风枪加热直到程序结束。参考图1.PSS、PS3传感管和P3B传感管。**

(10) Remove the moisture from the PS3 [4] and P3B [5] (with optional parts) as follows:
按以下步骤，执行PS3[4]和P3B[5](有选装零件)除水工作：

(a) Turn the vacuum relief valve [9] and blank-off valve [10] handles to the open position.
将真空释压活门[9]和断开活门[10]手柄转动到打开位。

**CAUTION: DO NOT LET CONTAMINATION BE INGESTED INTO THE VACUUM PUMP RELIEF PORT. IF THE PRESSURE READING IS NOT BELOW THE SPECIFIED THRESHOLD, THE PROCEDURE COULD NOT BE EFFECTIVE.**
**注意： 不要让污染物被吸入到真空泵释压口。如果压力读数不低于规定的阈值，该程序可能无效。**

(b) Start the vacuum pump [11] and let the pump motor get to the operating speed.
启动真空泵[11]并让泵马达达到工作速度。

**CAUTION: DO NOT CLOSE THE BLANK-OFF VALVE WHILE YOU REMOVE MOISTURE FROM THE PSS. IF THE BLANK-OFF VALVE IS CLOSED, MOISTURE WILL NOT BE REMOVED.**
**注意： 当从PSS去除水分时不要关闭断开活门。否则，水分将不会被去除。**

(c) Turn the valve handle of the vacuum relief valve [9] to the horizontal position to close it.
转动真空释压活门[9]的手柄到水平位置以关闭它。

(d) Make sure that sufficient vacuum is pulled on the PS3 [4] and P3B [5] manifolds by the vacuum pump [11].
确保真空泵[11]在PS3[4]和P3B[5]总管上抽到足够的真空。

NOTE: The vacuum read needs to be corrected based on the pressure altitude at which the pump is used. Refer toFigure Refer to Fig.3 Vacuum gauge corrections for altitude .

**注：** 真空读数需要基于泵在使用的压力海拔高度进行修正。参考图3.真空表海拔高度修正。

- If the airport elevation is below  5000 ft (  1524 m  ), make sure that the vacuum measurement is in the green-colored region on the vacuum pressure indicator.
如果机场海拔低于 5000 ft ( 1524 m )，确保真空测量值在真空压力指示器的绿色区域内。

- If the airport elevation is above 5000 ft ( 1524 m ), refer toFigure Refer to Fig .3 Vacuum gauge corrections for altitude to find the correct vacuum pressure me asurement.
-如果机场海拔高于 5000 ft ( 1524 m )，参考图3.真空表海拔高度修正，以找到正确的真空压力测量值。

- If the vacuum measurement is not correct, examine for leaks in the vacuum ho se, connections, and make sure the pump valves are in the correct position.
-如果真空测量值不正确，检查真空软管、连接是否渗漏，并确保泵活门处于正确位置。

(e) Operate the vacuum pump [11] for a minimum of 20 minutes.
真空泵[11]运行至少20分钟。

(f) After 20 minutes of operation, open the vacuum relief valve [9] to release the pre ssure.
运行20分钟后，打开真空释压活门[9]以释放压力。

(g) Turn OFF the vacuum pump [11].
关闭真空泵[11]。

(h) Remove the vacuum hose [12] end adapter from the PS3 pressure port [13] and P3B hose pressure port [14].
从PS3压力口[13]和P3B软管压力口[14]拆下真空管[12]端转接头。

(i) Blow clean dry Air Source-Compressed,Regulated,100 psi or Air Source - Nitroge n, Regulated, 100 psi (no more than 200 psi ( 1379 kPa )) in the PS3 sense line [ 2 ] and P3B sense line [ 3 ] for a minimum of 3 minutes. Continue until no moistur e comes out from the PS3 sense line [ 2 ] and P3B sense line [ 3 ] weep hole loc ated at the 4:00 o'clock position in the aft location of the fan compartment. Refer toFigure Refer to Fig.4 Location of Sense Line Weep Holes .
向PS3传感管[2]和P3B传感管[3]中吹清洁、干燥的100psi可调压缩空气源或100 psi可调氮气源(不超过 200 psi 1379 kPa ))至少3分钟。继续直到没有水汽从位于风扇舱后部4:00点钟位置的PS3传感管[2]和P3B传感管[3]渗水孔流出。参考图4.传感管渗水孔位置。

(j) Make sure that the PS3 sense line [2] and P3B sense line [3] weep holes are not blocked because of ice formation, damage, or contamination, and that there is a fr ee flow of air from the PS3 sense line [2] and P3B sense line [3] weep holes.
确保PS3传感管[2]和P3B传感管[3]渗水孔没有因结冰、损伤或污染而堵塞，并且有空气从PS3传感管[2]和P3B传感管[3]的渗水孔自由流出。

NOTE: For best results, compressed Air Source - Nitrogen, Regulate d, 100 psi is recommended.
**注:** 为了获得最佳效果，建议使用100 psi可调氮气源。

(11) Recovery the PS3 sense line [ 2 ] connection to PSS [1]. Refer to. PS3 line-Installati on,MPP C919-A-73-21-70-01A-720A-A .
恢复PS3传感管[2]连接到PSS[1]上。参考：PS3传感管 - 安装,MPP C919-A-73-21-70-01A-720A-A。

(12) Recovery the P3B sense line [ 3 ] connection to PSS [ 1 ]. Refer to P3B Line-Installat ion,MPP C919-A-73-21-75-01A-720A-A .
恢复P3B传感管[3]连接到PSS[1]上。参考:P3B传感管 - 安装, MPP C919-A-73-21-75-01A-720A-A。

(13) Refer toTable Table3 Vacuum pump kit details for the Vacuum pump kit(RC331931) details and refer toTable Table 4 Heat gun details for the Gun - Heat, 80-260℃ (180-500 ℉) details.
参考表3真空泵套件详情获取真空泵套件(RC331931)详情，并参考表4获取80-260℃(180-500℉)热风枪详情。

**Table 3 Vacuum pump kit details**
**表3 真空泵套件详情**

| SYSTEM OPERATION<br>系统操作 | QUANTITY<br>数量 | DESCRIPTION<br>描述 |
|---|---|---|
| Vacuum Pump 120V/60Hz.<br>真空泵120V/60Hz | 1 | Vacuum pump capable of 0.1 psi ( 6.8 mbar ) or lower with built-in gauge to 0 in ( 0 mm ) to 30 in ( 762 mm ) of Mercury (Hg).<br>真空泵压力为 0.1 psi(6.8 mbar)或更低，内置压力计，0in(0mm) - 30in(762mm)汞柱。<br>2-stage rotary vane super vac with gauge. (1/4″ and 3/8″ male flare fitting supplied with pump).<br>带仪表的2级旋转叶片超级真空吸尘器。（随泵提供的1/4″和3/8″外螺纹扩口接头）。 |
| Pump oil.<br>泵滑油 | 1 | Vacuum oil (12 Qts.).<br>真空滑油(12夸脱) |
| Start/Bleed Valve.<br>启动/放气活门 | 1 | Ball Valve 3/8″ sea male FI. X female FI.<br>球阀3/8″外螺纹FI.X内螺纹FI. |
| Breather vent.<br>通气口 | 1 | Sintered metal filter 40 microns.<br>烧结金属过滤器40微米。 |
| Female NPT to Female NPT coupler.<br>内螺纹NPT至内螺纹NPT接头。 | 0.1 | Brass coupler (package of 10) 3/8″ female NPT to 3/8″ female NPT.<br>黄铜耦合器（10件套）3/8″内螺纹NPT转3/8” 内螺纹NPT。 |
| Smoke eliminator filter.<br>烟雾消除过滤器 | 1 | Smoke eliminator filter for super evac pump.<br>适用于SuperEvac泵的烟雾消除过滤器。 |
| Vacuum hose.<br>真空软管 | 1 | Swagelok PTFE hose with fiber braid, 12’ in length, pump connector is 3/8″ Swagelok tube fitting, engine control connector is 3/8″ JIC (AN) fitting.<br>带纤维编织物的Swagelok PTFE软管，长度为12英寸，泵接头为3/8″Swagelok套管接头，发动机控制接头为3/8″JIC（AN）接头。 |
| Pump fitting.<br>泵接头 | 1 | Swagelok stainless steel tube fitting, male connector, 3/8 in. Tube OD x 1/8 in. Male NPT. This fitting replaces the 1/4″ SAE fitting on the pump.<br>Swagelok不锈钢卡套管接头，外螺纹接头，3/8 in.卡套管外径x 1/8 in.外螺纹NPT。此接头更换了泵上的1/4″SAE接头。 |

**Table 4 Heat gun details**
**表4 热风枪详情**

| Item<br>项目 | Description<br>描述 | Specification<br>规范 | Optional/Required Accessories<br>可选/必需配件 |
| --- | --- | --- | --- |
| Heat Gun.<br>热风枪 | Steinel HG 2310 LCD three-stage professional heat gun or equivalent.<br>Steinel HG 2310 LCD三级专业热风枪或同等产品。 | Heat gun features LOC lockable override control (trademark) of temperature and airflow output settings. The LCD display enables temperature selection in increments of 10 F ( 5.55 ℃ ) degrees.<br>热风枪具有LOC可锁定的温度和气流输出设置超控（商标）功能。LCD显示屏可实现以 10 F ( 5.55 ℃ ) 为增量的温度选择。 | Two voltage versions, 120V USA and 230V Euro.<br>两种电压版本，120V USA和230V Euro。 |

C. Close-Up
结束

(1) Do the access job close up after maintenance on the fan area. Refer to Power plant ( Fan area)-Close after access procedure, MPP C919-A-71-00-00-03A-740A-A.
维护结束后，执行风扇区域关闭工作。参考：动力装置（风扇区域） - 接近程序后关闭,MPP C919-A-71-00-00-03A-740A-A 。

(2) Do one of the tests that follow: .
执行下列程序之一：

NOTE: The idle test (Step Step C.(2).(a). ) will show failed if PS3 or P3B faults are detected during the test. The idle leak test (Step Step C.(2).(b). ) will require to do a check of the fault history.

**注：** 如果在测试过程中检测到PS3或P3B故障，慢车测试(步骤C.(2).(a).)将显示失败。慢车渗漏测试(步骤C.(2).(b).)将需要检查故障历史记录。

(a) Do the idle test (menu mode). Refer to Power plant-Idle test,MPP C919-A-71-00-00-01A-345C-A .
执行慢车测试(菜单模式)。参考：动力装置 - 慢车测试,MPP C919-A-71-00-00-01A-345C-A。

(b) Do the idle leak test. Refer to Power plant-Idle leak check,MPP C919-A-71-00-00-01A-364C-A .
执行慢车渗漏测试。参考：动力装置 - 慢车渗漏测试,MPP C919-A-71-00-00-01A-364C-A。

(3) Remove all the WARNING NOTICE(s).
拆下全部警告牌。

(4) Remove all tools, equipment, and unwanted materials from the work area.
从工作区域移除所有的工具、设备和外来物。